

Continuous Dynamics

smooth and non smooth motion

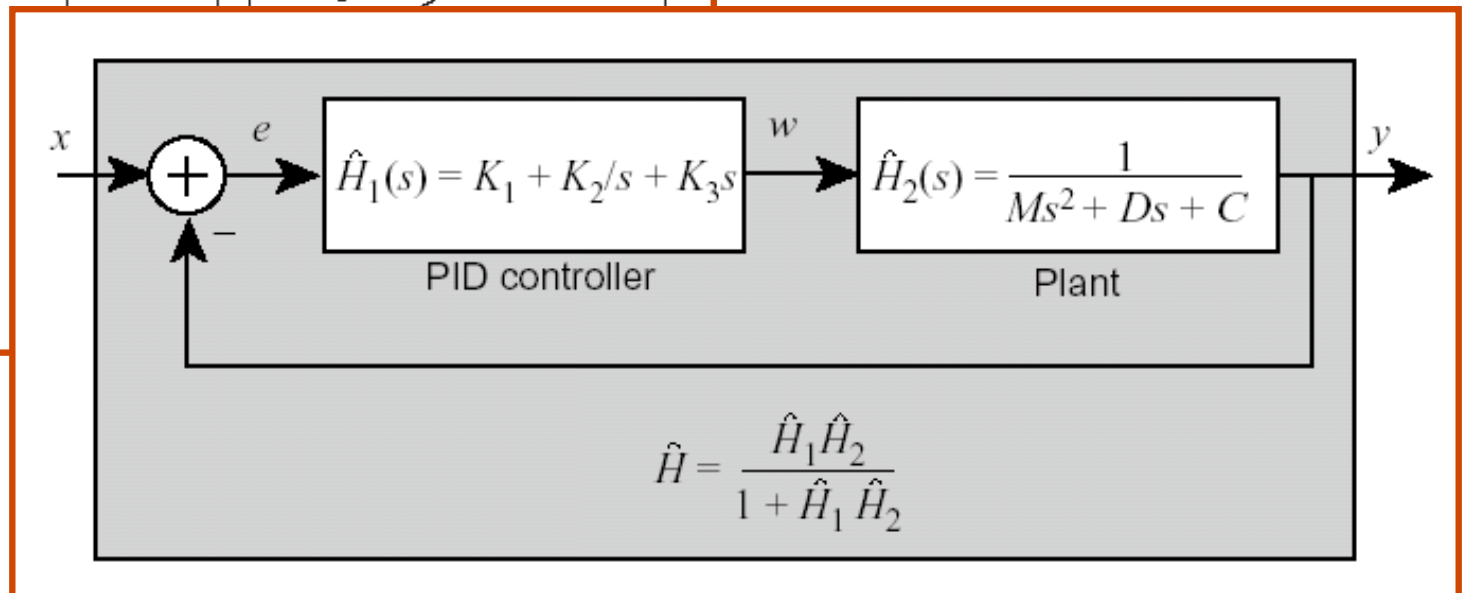
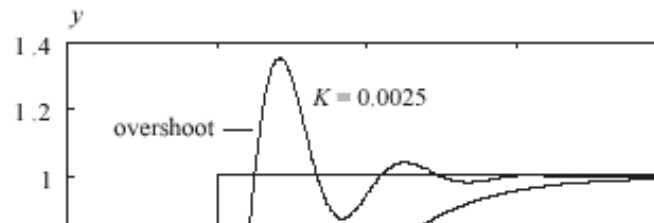
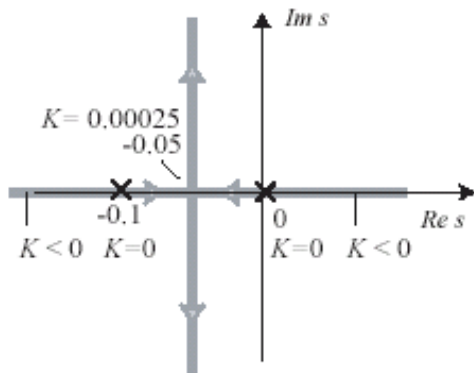
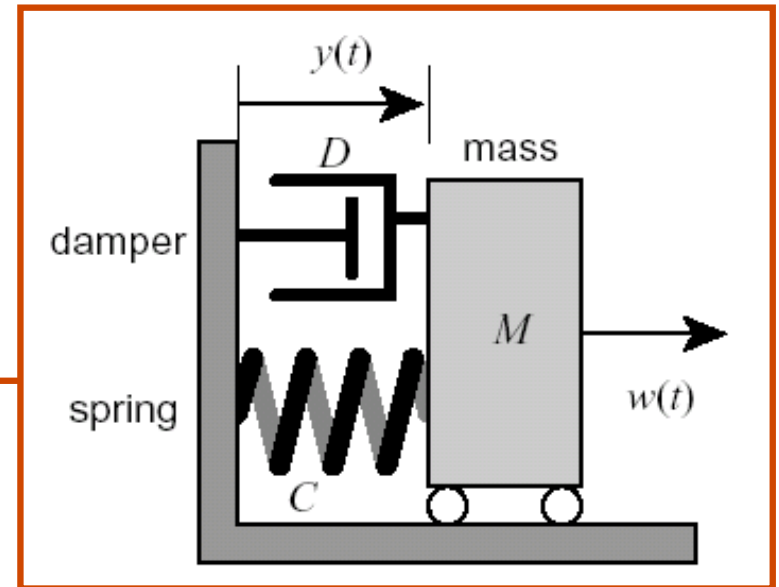
- smooth: notion of linearity, time invariance, and continuity
- non smooth: abrupt, discrete, one smooth motion translates into other smooth motion
- joint smooth and non smooth motion — hybrid systems
- combination of discrete and continuous behaviors is close to modeling of cyber and physical system.

Modeling a Dynamical System

- Modeling Method: ordinary differential equations(ODEs).....
- Modeling Language: LabVIEW, Simulink(from matlab)
- Properties of such models: linearity, time invariance, stability,

Modeling of Continuous Dynamics

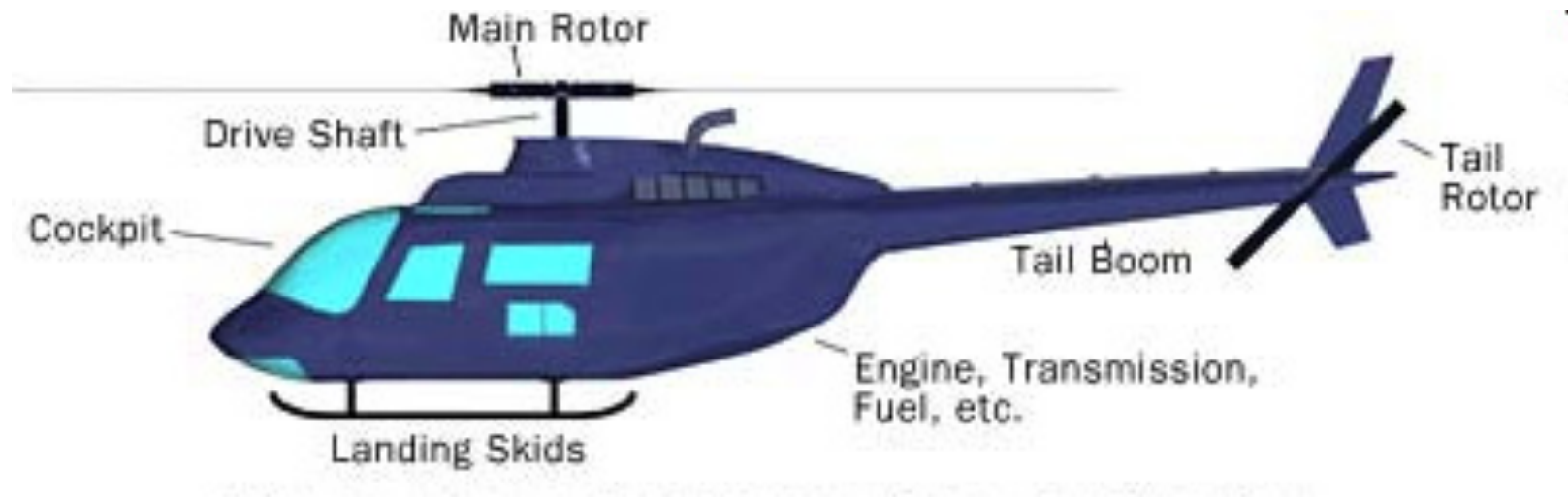
Ordinary differential equations, Laplace transforms, feedback control systems, stability analysis, robustness analysis, ...



Continuous Dynamics

- Classical mechanics
- smooth motion
 - ✦ linearity
 - ✦ time invariance
 - ✦ continuity
- modeling with tool
 - ✦ matlab

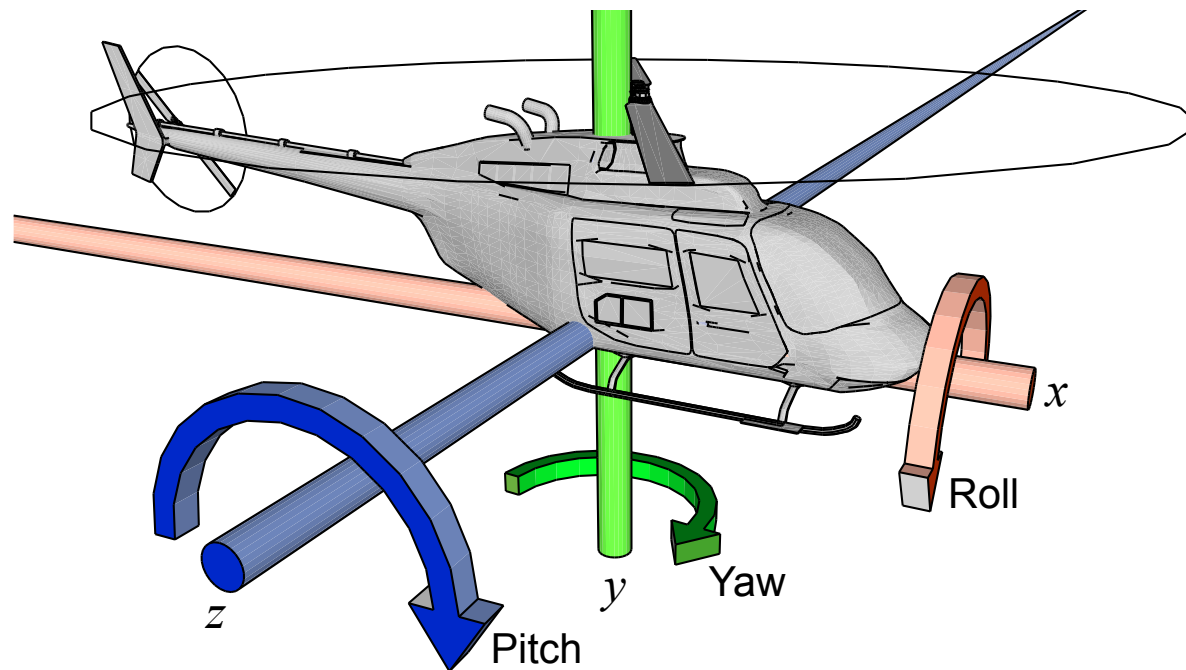
An Example: Modeling Helicopter Dynamics



Modeling Physical Motion

Six degrees of freedom:

- Position: x , y , z
- Orientation: roll (橫滾), yaw (偏航), pitch(俯仰)



Notation

- Position is given by three functions:

$$x: \mathbb{R} \rightarrow \mathbb{R}$$

$$y: \mathbb{R} \rightarrow \mathbb{R}$$

$$z: \mathbb{R} \rightarrow \mathbb{R}$$

- where the domain represents time and
- the codomain represents either distance along an axis.
- Collecting into a vector:

$$\mathbf{x}: \mathbb{R} \rightarrow \mathbb{R}^3$$

Position at time $t \in \mathbb{R}$ is $\mathbf{x}(t) \in \mathbb{R}^3$

Notation

Velocity

$$\dot{\mathbf{x}}: \mathbb{R} \rightarrow \mathbb{R}^3$$

is the derivative, $\forall t \in \mathbb{R}$,

$$\dot{\mathbf{x}}(t) = \frac{d}{dt}\mathbf{x}(t)$$

Acceleration $\ddot{\mathbf{x}}: \mathbb{R} \rightarrow \mathbb{R}^3$ is the second derivative,

$$\ddot{\mathbf{x}} = \frac{d^2}{dt^2}\mathbf{x}$$

Force on an object is $\mathbf{F}: \mathbb{R} \rightarrow \mathbb{R}^3$.

Newton's Second Law

Newton's second law

$$\mathbf{F}(t) = M\ddot{\mathbf{x}}(t)$$

M is the mass of the object

$$\forall t > 0, \quad \dot{\mathbf{x}}(t) = \dot{\mathbf{x}}(0) + \int_0^t \ddot{\mathbf{x}}(\tau) d\tau$$

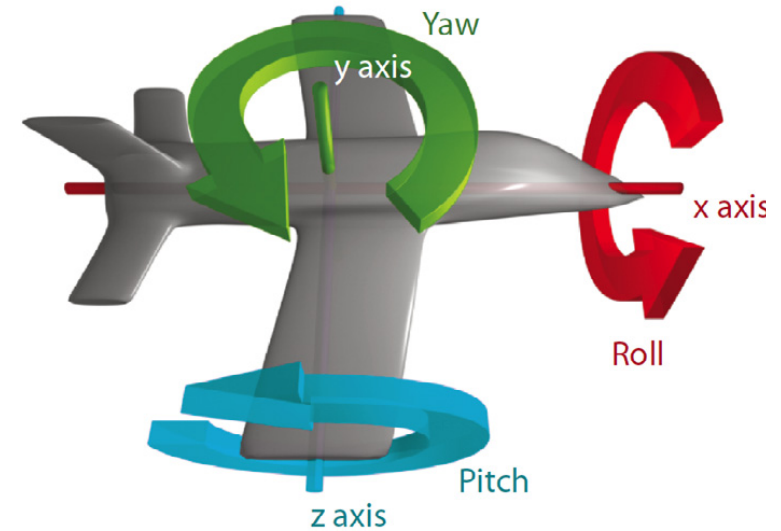
$$\forall t > 0, \quad \dot{\mathbf{x}}(t) = \dot{\mathbf{x}}(0) + \frac{1}{M} \int_0^t \mathbf{F}(\tau) d\tau,$$

Newton's Second Law

$$\begin{aligned}\mathbf{x}(t) &= \mathbf{x}(0) + \int_0^t \dot{\mathbf{x}}(\tau) d\tau \\ &= \mathbf{x}(0) + t\dot{\mathbf{x}}(0) + \frac{1}{M} \int_0^t \int_0^\tau \mathbf{F}(\alpha) d\alpha d\tau,\end{aligned}$$

Orientation

- Orientation: $\theta: \mathbb{R} \rightarrow \mathbb{R}^3$
- Angular velocity: $\dot{\theta}: \mathbb{R} \rightarrow \mathbb{R}^3$
- Angular acceleration: $\ddot{\theta}: \mathbb{R} \rightarrow \mathbb{R}^3$
- Torque: $\mathbf{T}: \mathbb{R} \rightarrow \mathbb{R}^3$



$$\dot{\theta}(t) = \begin{bmatrix} \dot{\theta}_x(t) \\ \dot{\theta}_y(t) \\ \dot{\theta}_z(t) \end{bmatrix} = \begin{bmatrix} \text{roll} \\ \text{yaw} \\ \text{pitch} \end{bmatrix}$$

angular acceleration

$$\mathbf{T}(t) = \frac{d}{dt} (\mathbf{I}(t) \dot{\boldsymbol{\theta}}(t))$$

- $\mathbf{T}$ is the torque (扭力) vector in three axes.
- $\mathbf{I}(t)$ is the moment of inertia tensor(惯性张量) of the object. a 3×3 matrix that depends on the geometry (几何形状) and orientation (方位) of the object.

$$\begin{bmatrix} T_x(t) \\ T_y(t) \\ T_z(t) \end{bmatrix} = \frac{d}{dt} \left(\begin{bmatrix} I_{xx}(t) & I_{xy}(t) & I_{xz}(t) \\ I_{yx}(t) & I_{yy}(t) & I_{yz}(t) \\ I_{zx}(t) & I_{zy}(t) & I_{zz}(t) \end{bmatrix} \begin{bmatrix} \dot{\theta}_x(t) \\ \dot{\theta}_y(t) \\ \dot{\theta}_z(t) \end{bmatrix} \right)$$

simple example

$$\mathbf{T}(t) = I\ddot{\boldsymbol{\theta}}(t), \quad \text{where a constant scalar } I \text{ for a spherical}$$

Rotational velocity is the integral of acceleration

$$\dot{\boldsymbol{\theta}}(t) = \dot{\boldsymbol{\theta}}(0) + \int_0^t \ddot{\boldsymbol{\theta}}(\tau) d\tau, \quad \dot{\boldsymbol{\theta}}(t) = \dot{\boldsymbol{\theta}}(0) + \frac{1}{I} \int_0^t \mathbf{T}(\tau) d\tau.$$

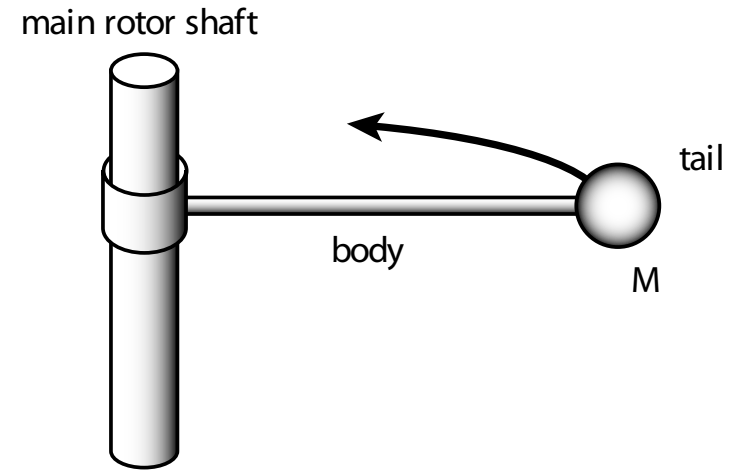
$$\begin{aligned} \boldsymbol{\theta}(t) &= \boldsymbol{\theta}(0) + \int_0^t \dot{\boldsymbol{\theta}}(\tau) d\tau \\ &= \boldsymbol{\theta}(0) + t\dot{\boldsymbol{\theta}}(0) + \frac{1}{I} \int_0^t \int_0^\tau \mathbf{T}(\alpha) d\alpha d\tau \end{aligned}$$

example

- model-order reduction: reduce the number of dimension that are consider
- a moving vehicle on a flat surface
- orientation only consider y axis

example

- A helicopter with two rotors
 - ✦ position is fixed.
 - ✦ the tail rotor counteracts the main rotor's spinning
 - ✦ remains vertical, pitch and roll are zero, only has yaw
- consider the tail rotor

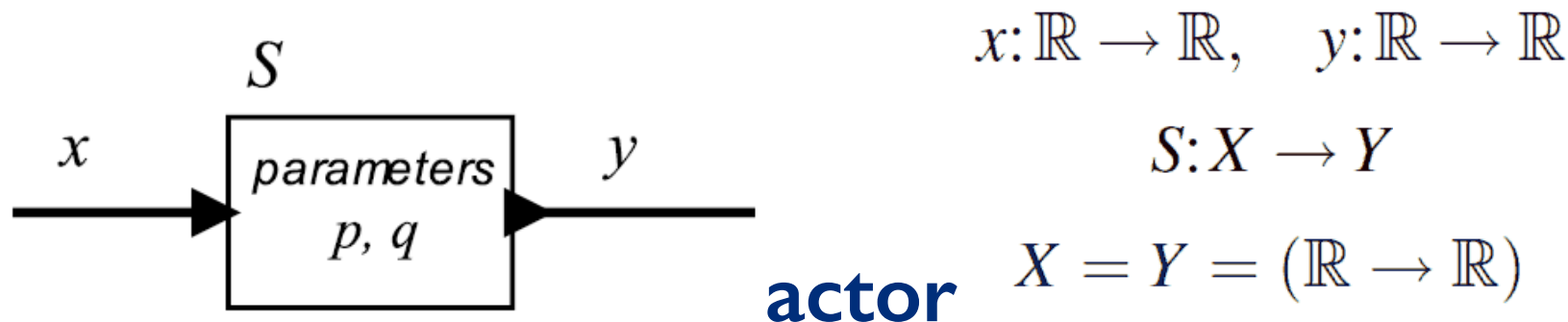


$$\ddot{\Theta}_y(t) = T_y(t)/I_{yy}.$$

$$\dot{\Theta}_y(t) = \dot{\Theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau.$$

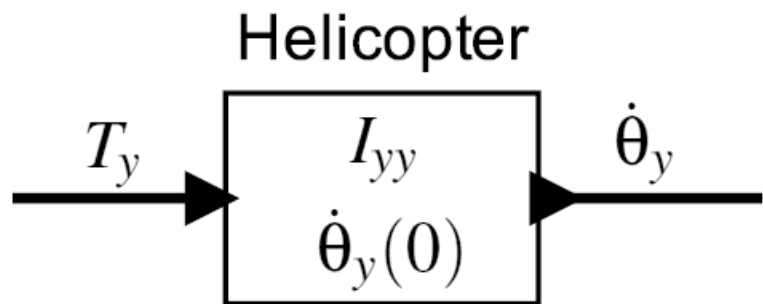
Actor Model of Systems

- A *system* is a function that accepts an input *signal* and yields an output signal.
- In a large system, it is a *component*
- The domain and range of the system function are sets of signals, which themselves are functions.
- *Parameters* may affect the definition of the function S .



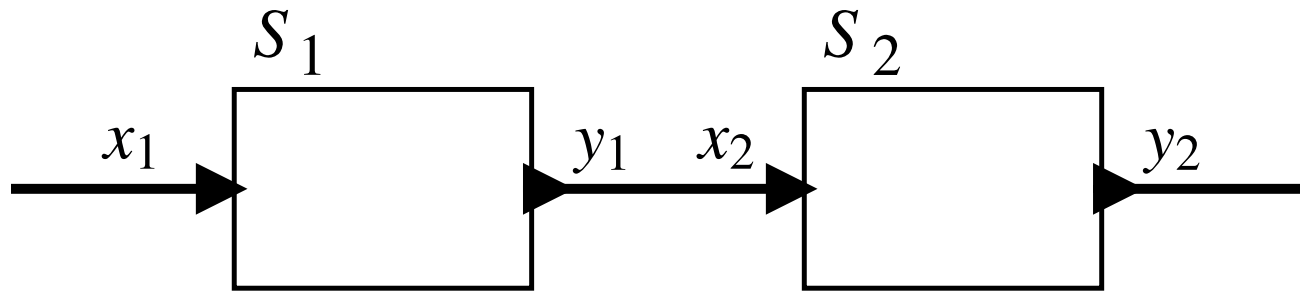
Actor model of the helicopter

- Input is the net torque of the tail rotor and the top rotor.
- Output is the angular velocity around the y axis.
- Parameters of the model are shown in the box.
- The input and output relation is given by the equation to the right.



$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau$$

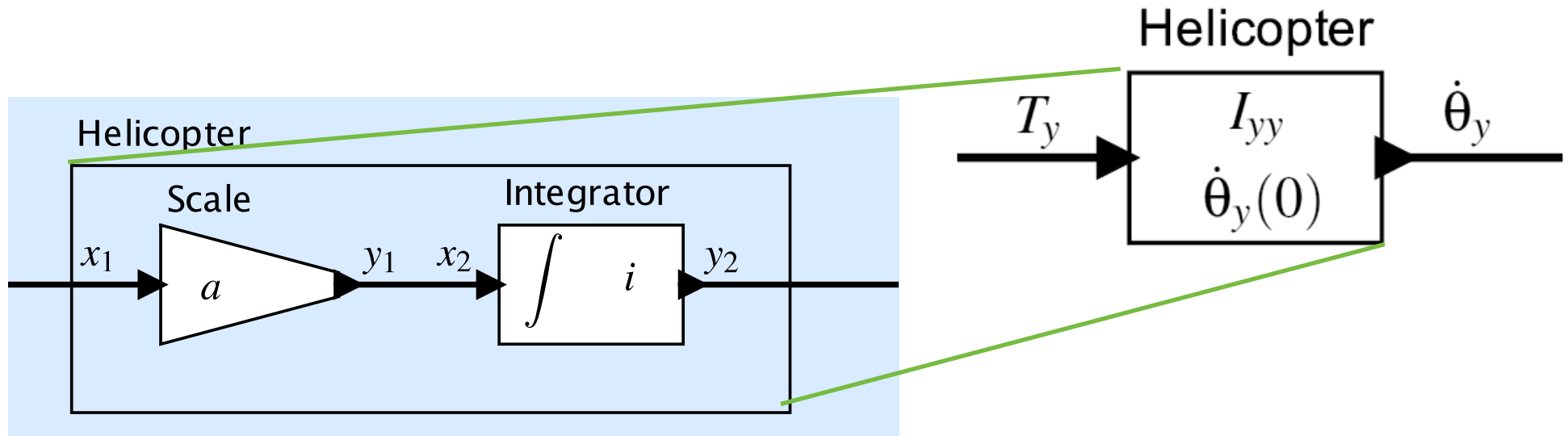
Actor models is composable



- Cascade composition: the “wire” between the output of S_1 and the input of S_2 means precisely that $y_1 = x_2$,

$$\forall t \in \mathbb{R}, y_1(t) = x_2(t).$$

helicopter example

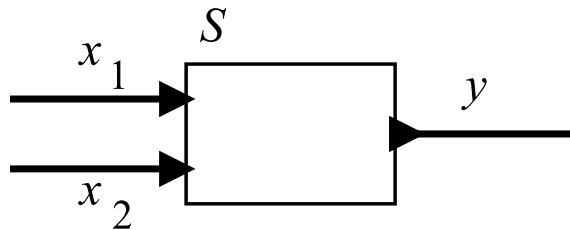


$$\forall t \in \mathbb{R}, \quad y_1(t) = ax_1(t).$$

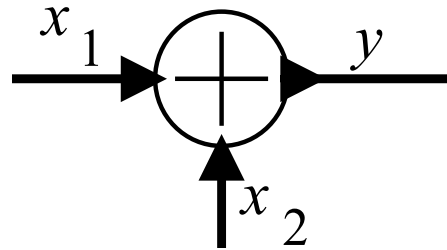
$$\forall t \in \mathbb{R}, \quad y_2(t) = i + \int_0^t x_2(\tau) d\tau.$$

$$a = 1/I_{yy} \text{ and } i = \dot{\theta}_y(0)$$

Actor models with multiple inputs

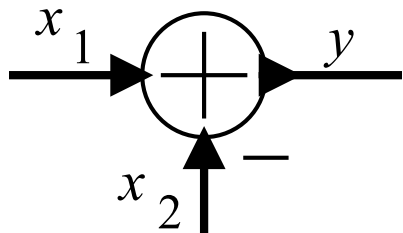


- two input signals and one output signal



- adder

$$\forall t \in \mathbb{R}, y(t) = x_1(t) + x_2(t).$$



- one of the inputs will be subtracted rather than added

$$S : (\mathbb{R} \rightarrow \mathbb{R})^2 \rightarrow (\mathbb{R} \rightarrow \mathbb{R})$$

$$\forall t \in \mathbb{R}, \forall x_1, x_2 \in (\mathbb{R} \rightarrow \mathbb{R}),$$

$$(S(x_1, x_2))(t) = y(t) = x_1(t) - x_2(t).$$

Properties of System

- causality(因果性)
- memoryless (无记忆性)
- linearity (线性性)
- time invariance (时不变性)
- stability (稳定性)

causality

- its output depends only on current and past inputs.
- a continuous-time signal $x : \mathbb{R} \rightarrow A$
- $x|_{t \leq \tau}$ represent a function called the restriction in time
$$x|_{t \leq \tau}(t) = x(t)$$
- if x is an input to a system, then $x|_{t \leq \tau}$ is the “current and past inputs” at time τ .

causality

- This system is causal if for all $x_1, x_2 \in X$ and $\tau \in \mathbb{R}$,

$$x_1|_{t \leq \tau} = x_2|_{t \leq \tau} \Rightarrow S(x_1)|_{t \leq \tau} = S(x_2)|_{t \leq \tau}$$

- All systems we have considered so far are causal

strictly causal

- for all $x_1, x_2 \in X$ and $\tau \in \mathbb{R}$,
$$x_1|_{t < \tau} = x_2|_{t < \tau} \Rightarrow S(x_1)|_{t \leq \tau} = S(x_2)|_{t \leq \tau}$$
- for two possible inputs x_1 and x_2 that are identical up to (not including) time τ , outputs are identical up to (including) time τ .
 - The Integrator actor: strictly causal;
 - the adder: not strictly causal, but causal;
- Strictly causal actors are useful for constructing **feedback** systems.

Memoryless Systems

- $S : X \rightarrow Y$, where $X = A^{\mathbb{R}}$ and $Y = B^{\mathbb{R}}$
- exist a function $f : A \rightarrow B$ such that for all $x \in X$,

$$(S(x))(t) = f(x(t))$$

- output is decided by input at time t .
- example:the adder is the memoryless system, but the Integrator is not.
- if a system is strictly causal and memoryless then its output is constant for all inputs (?)

Linearity and Time Invariance

- superposition property (叠加性质) :

for a system $S: X \rightarrow Y$,

$$\forall x_1, x_2 \in X \text{ and } \forall a, b \in \mathbb{R},$$

$$S(ax_1 + bx_2) = aS(x_1) + bS(x_2).$$

- A system is linear if it satisfies the superposition property

- example

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau.$$

$$\forall t \in \mathbb{R}, \quad y_2(t) = i + \int_0^t x_2(\tau) d\tau.$$

- cascade of any two linear actors is linear.
- the Scale actor, the adder are always linear

Linearity and Time Invariance(LTI)

- delay actor: $D_\tau : X \rightarrow Y$, where X and Y are sets of continuous-time signals, be defined by

$$\forall x \in X \text{ and } \forall t \in \mathbb{R}, (D_\tau(x))(t) = x(t - \tau).$$

τ is a parameter of the delay actor

- A system $S : X \rightarrow Y$ is time invariant if

$$\forall x \in X \text{ and } \forall \tau \in \mathbb{R}, S(D_\tau(x)) = D_\tau(S(x)).$$

- linear time-invariant system (LTI): linear and time-invariant

- A major objective in modeling physical dynamics is to choose an LTI model

Stability

- **bounded-input bounded-output stable (BIBO stable or just stable)** if the output signal is bounded for all input signals that are bounded.

- example

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau.$$

- ♦ Let the input be $T_y = u$, where u is the **unit step**

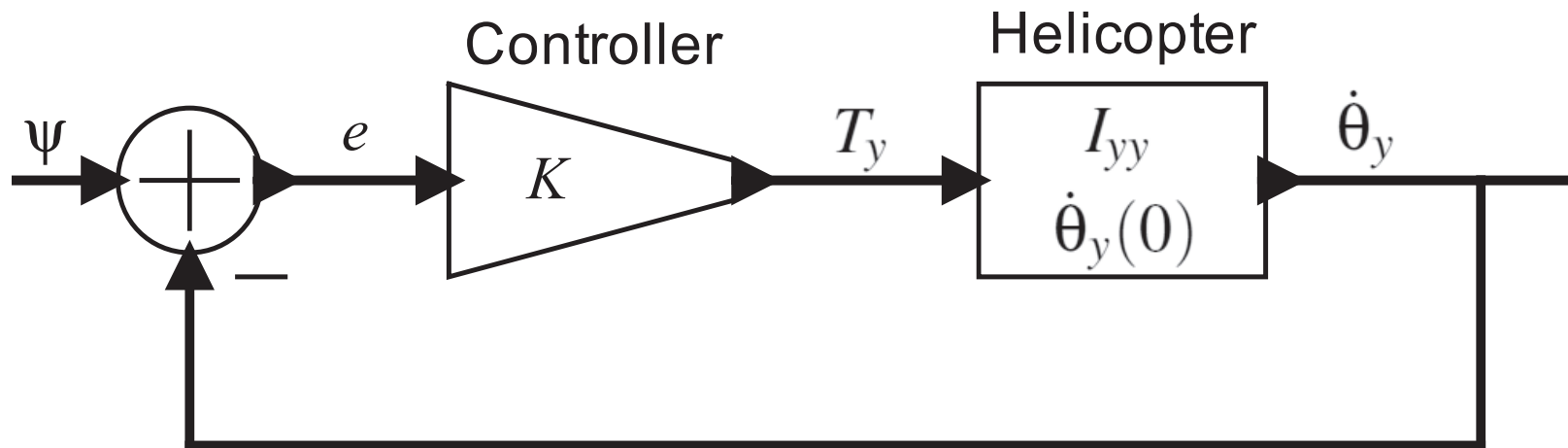
$$\forall t \in \mathbb{R}, \quad u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

- ♦ This input is clearly bounded.

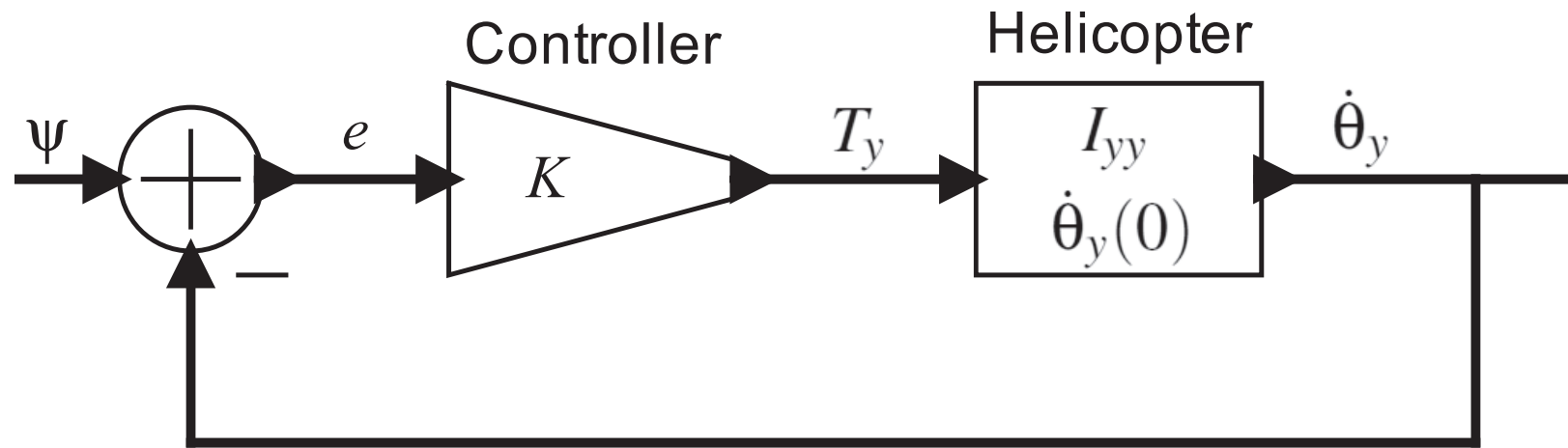
- ♦ the output grows without bound.

2.4 Feedback Control

- A system with **feedback** has directed cycles, where an output from an actor is fed back to affect an input of the same actor.

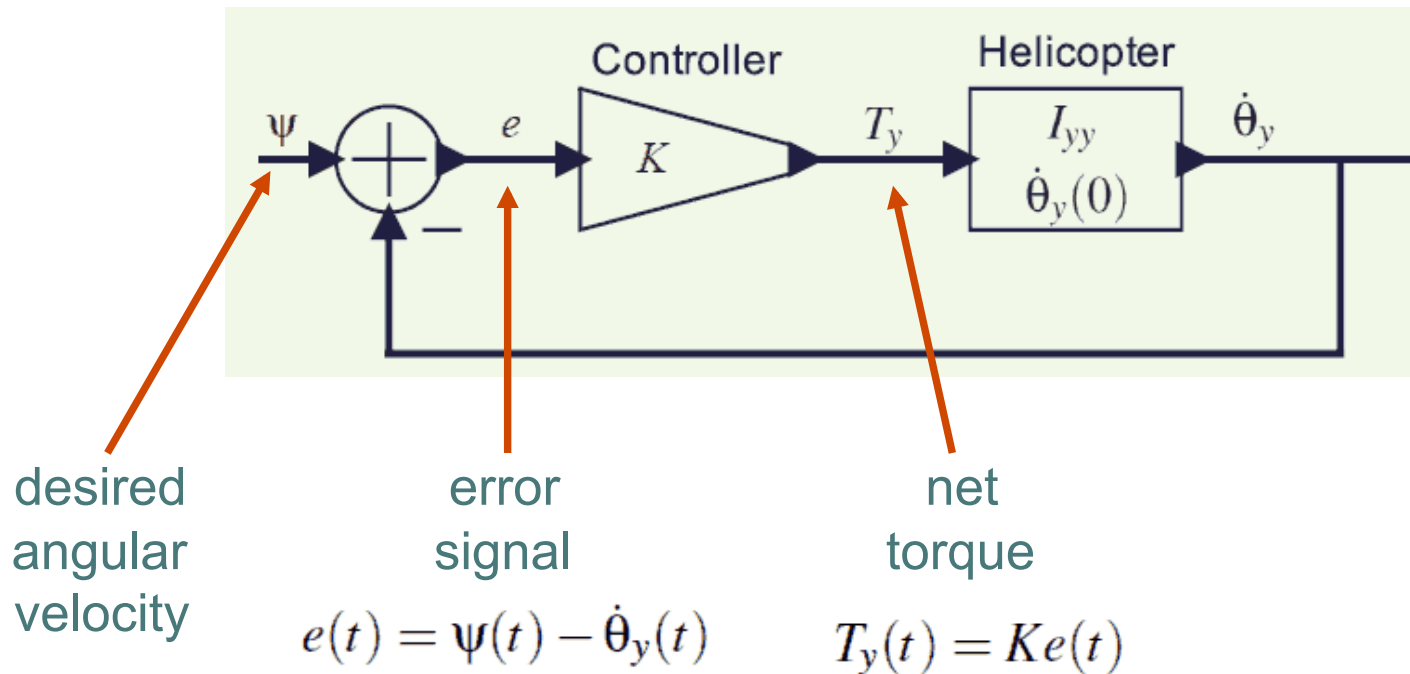


Feedback Control



- comparison desired behavior (ψ) and actual behavior ($\dot{\theta}_y$)
- the corresponding correction signal (T_y) should compensate to reduce future error
- at least one strictly causal system

Proportional controller

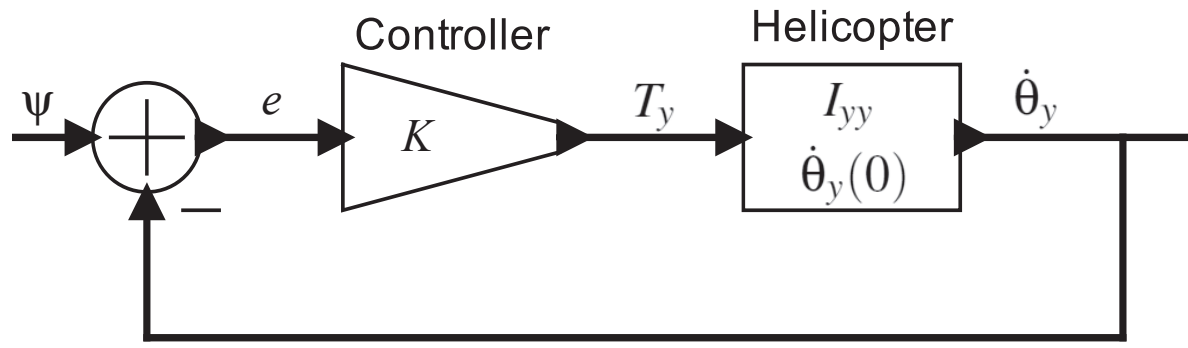


$$\begin{aligned}\dot{\theta}_y(t) &= \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau \\ &= \dot{\theta}_y(0) + \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau,\end{aligned}$$

Note that the angular velocity appears on both sides, so this equation is not trivial to solve.

- control signal is proportional to the error.

example1:Proportional control feedback



$$e(t) = \psi(t) - \dot{\theta}_y(t), \quad T_y(t) = Ke(t).$$

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau,$$

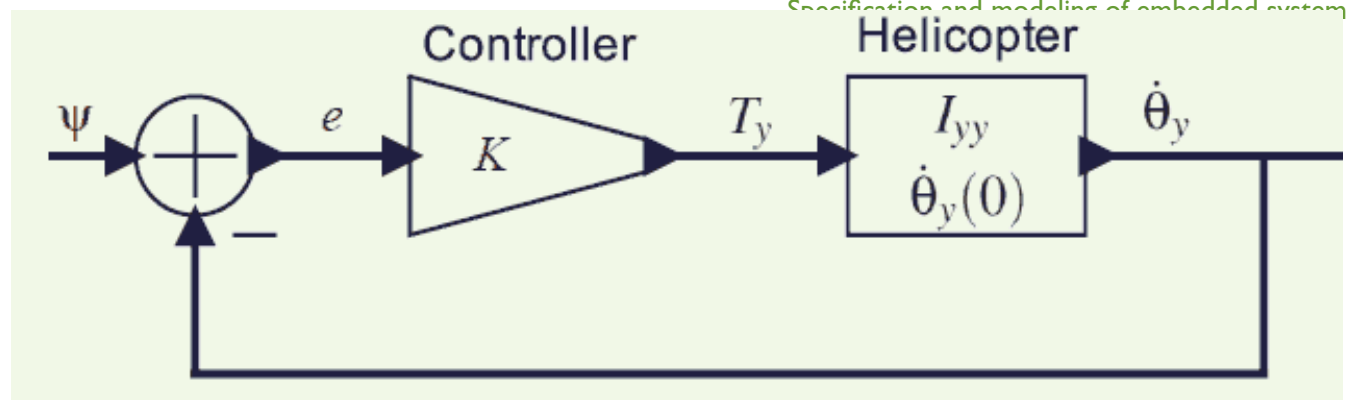
assuming that $\psi(t) = 0$

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau.$$

the solution is

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) e^{-Kt/I_{yy}} u(t).$$

example2



$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau$$

Assume that helicopter is initially at rest,

$$\dot{\theta}(0) = 0,$$

and that the desired signal is

$$\psi(t) = au(t)$$

for some constant a .

By calculus (see notes), the solution is

$$\dot{\theta}_y(t) = au(t)(1 - e^{-Kt/I_{yy}})$$

$$\begin{aligned} &= \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau \\ &= \frac{K}{I_{yy}} \int_0^t a d\tau - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau \\ &= \frac{Kat}{I_{yy}} - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau. \end{aligned}$$

example3

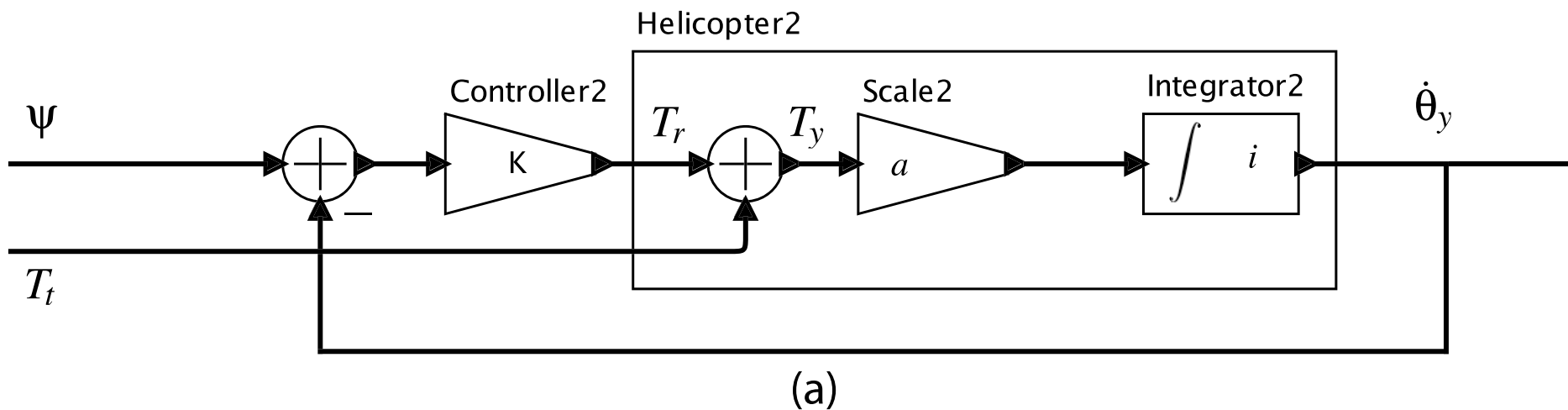
- the net torque T_y is the sum of the torque T_t due to the friction of the top rotor and the torque T_r due to the tail rotor,

$$\forall t \in \mathbb{R}, T_y(t) = T_t(t) + T_r(t).$$

where T_t is determined by the rotation to an altitude.

- design a control system that controls T_r

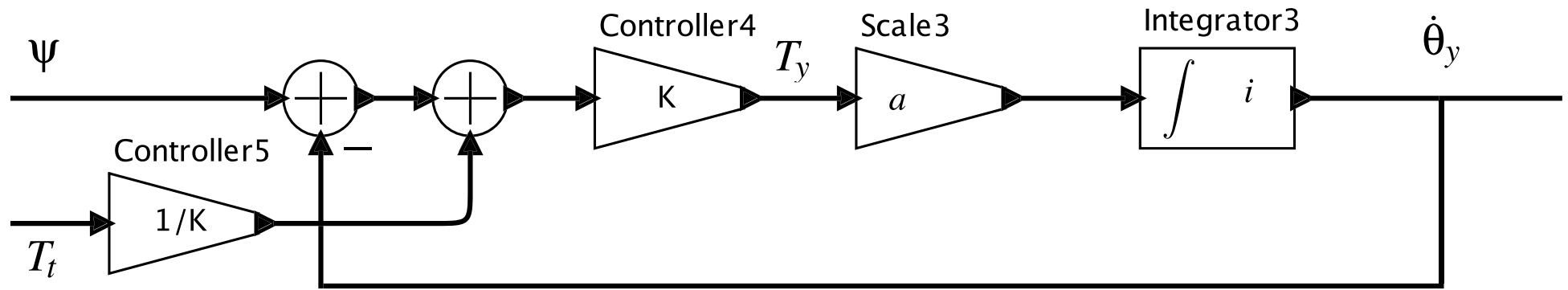
example3 (cont')



- Helicopter model with separately controlled torques for the top and tail rotors.

$$\dot{\theta}(0) = 0, \quad T_t = bu(t)$$

example3 (cont')

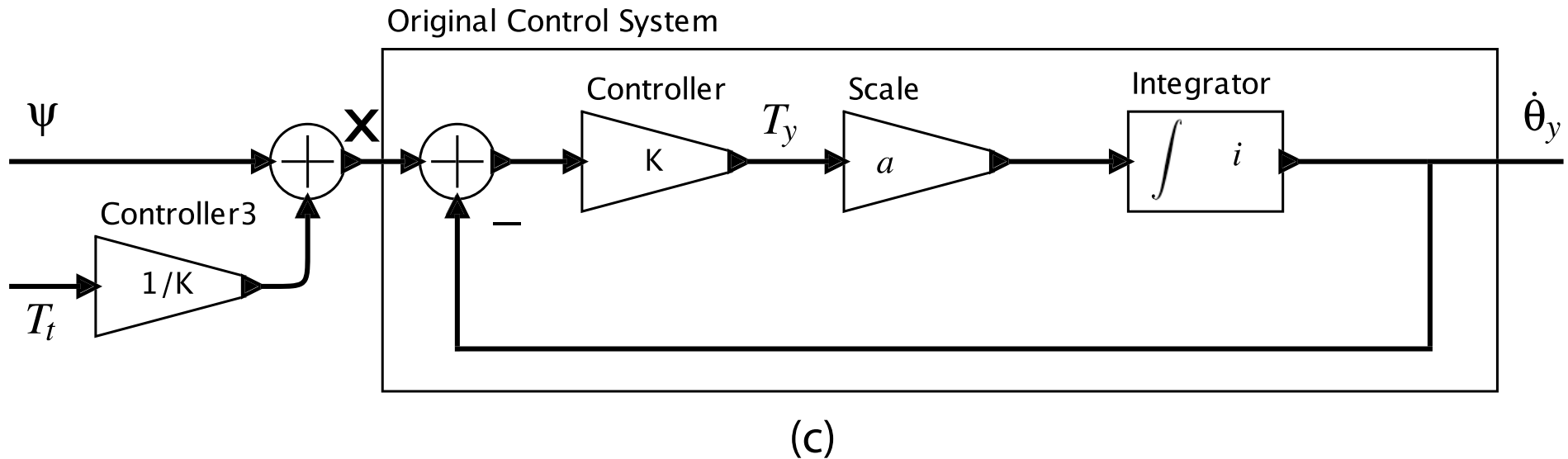


(b)

- Transformation to an equivalent model (assuming $K > 0$).
- reason is

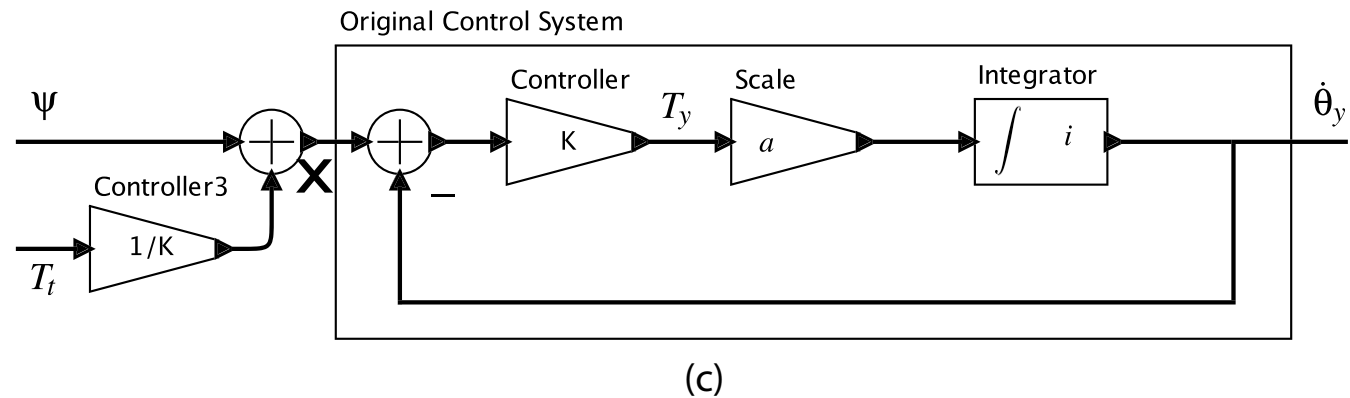
$$K a_1 + a_2 = K(a_1 + a_2/K).$$

example3 (cont')



- addition is commutative.
- Further transformation to an equivalent model that we can use to understand the behavior of the controller.

example3 (cont')



Assuming $\psi(t)=0$.

Then the input to the original control system will be

$$x(t) = \psi(t) + T_t(t)/K = (b/K)u(t).$$

the solution is

$$\dot{\theta}_y(t) = (b/K)u(t)(1 - e^{-Kt/I_{yy}}).$$

summary

- methods of modeling physical dynamics
 - ordinary differential equation
 - actor model---software modeling
- properties of system
 - linearity
 - time invariance
 - stability
- feedback control system